Abu Saleh

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Profile

Research and Development-focused Master's student in Software Engineering and Machine Learning with a strong passion to improve the design of intelligent systems and software. Currently exploring Small Language Models (SLMs) and Agentic AI to automate engineering working pipelines. I am seeking a junior position focused on foundational AI (LLMs) and other AI/ML domains. View the results of my coursework here.

Education

Master's in Engineering of Data-intensive Intelligent Software Systems
Åbo Akademi University, Finland & University of L'Aquila, Italy
Aug 2024 – Oct 2026
Finland & Italy

- **Year 1:** Master of Science(Technology) — foundations in data-intensive systems and **core machine learning**.
- **Year 2:** Master of Science: Software Engineering — Specialization in **Model Driven Machine Learning**.
- **Focus Areas:** Machine Learning, Large Language Models, Small Language Models, Multi-Agent Systems, Model-Driven Engineering, Software Engineering for Autonomous Systems, Software Architectures.

Bachelor's in Computer Science and Engineering
Daffodil International University
May 2019 – Dec 2023
Dhaka, Bangladesh

- GPA: 3.95/4.0, Highest Distinction, Merit-based Scholarship, Teaching Assistant, Programming Trainer.
- **Coursework:** Data Structures, Algorithms, Data Mining, Software Engineering, Computer Architecture.

Skills

- **ML & Deep Learning**: Python, PyTorch, Scikit-learn, Torchaudio, Computer Vision, Forecasting.
- **Generative AI**: LLMs/SLMs, RAG, MCP, Multi-Agent Systems, Vector Databases, LangChain, Google-Adk.
- **MLOps & Reproducibility**: MLflow, Optuna, Docker, Git, Azure ML, Experiment Tracking, Model Versioning.
- **Data & Analytics**: SQL, PostgreSQL, Databricks, Time-Series Analysis, Data Quality, SHAP.
- **Software Engineering**: REST APIs, Django REST Framework, Flask, React, PostgreSQL, Pytest.
- **Low-Level & Edge**: C/C++, Model-to-C Translation, Edge Inference Optimization, Model Distillation.

Professional Experiences

AI Engineer – Test Automation (Part-time)
u-blox
Jun 2026 – Present
Tampere, Finland

- Designed a **multi-agent LLM pipeline** (orchestrator, migration, verification and reporting agents) that ports GNSS performance KPIs from a legacy Perl test framework to Python/Robot Framework with verification.
- Built **2 MCP servers in Python (21 tools)** that give agents access to a test-results database, a log archive and SharePoint Excel (Microsoft Graph), with guardrails such as host allow-lists, read-only SQL and certificate auth.
- Migrated **˜40 KPI algorithms and 70+ automated analysis tests**, checking agent output against legacy results on real receiver logs (e.g. **64/64 values matched** across 8 devices).
- Cut MCP query latency **˜10× (83 s → ˜7 s)** and data transfer **˜1000× (276 MB → 273 KB)** using server-side filtering and parameterized SQL, which also kept agent context small and reduced **token usage by 13%**.
- **Skills:** Python, LLM agents, MCP, tool calling, prompt engineering, async APIs (httpx), Robot Framework.

Data Scientist Intern (Part-time)
Montel
Mar 2026 – Oct 2026
Sweden (Remote)

- Built **mFRR capacity-price forecasting pipelines** (LightGBM, Darts) for Finland and Sweden, producing 72-hour UP/DOWN forecasts across 5 bidding zones and **reducing forecast error by 7%**.
- Developed **regime-switch model** for Swedish prices, detecting **82% of price spikes** (37/45) in historical replays.
- Productionized **Airflow pipelines** for multi-source energy data, including a forecast-correction workflow across **22 European bidding zones**, and wrote **˜15 product data specifications** for the EnAppSys platform.

- **Associated Skills:** Python, LightGBM, Darts, time-series forecasting, Airflow, pytest, Git.

Audio Junior Researcher (Intern) May 2025 – Aug 2025
Huawei Technologies Oy Tampere, Finland

- Optimized **VGG-based CNNs** for acoustic scene classification, improving accuracy by **8%** via FFT tuning.
- Refactored legacy workflows into **modular Python pipelines** enabling systematic ablation studies.
- Translated **VGGish model layers to C** for low-latency edge inference with no loss in performance.
- **Associated Skills:** PyTorch, CNNs, Audio Signal Processing, FFT, C, MLflow, Git.

Research Intern – Time-Series Sequence Modeling Oct 2024 – May 2025
Elisa Oy Turku, Finland

- Built and evaluated **forecasting models** (LightGBM, TFT) for highly volatile energy markets (**MAE of 17–21**).
- Designed **end-to-end ML pipelines** with 90+ engineered temporal features and **sliding-window backtesting**.
- Improved model transparency using **SHAP-based explainability** to analyse feature impact and reliability.
- **Associated Skills:** Time-Series Forecasting, LightGBM, Temporal Fusion Transformers, SHAP, Optuna.

Software Engineer Sept 2023 – Apr 2024
Enosis Solutions Dhaka, Bangladesh

- Built backend APIs and **data-driven dashboards** supporting analytical workflows and business reporting.
- Designed task management features using DFS topological sort and implemented **Role-Based Access Control**.
- **Associated Skills:** React.js (UI/UX), Data Visualization, Django Rest Framework, SQLAlchemy, PostgreSQL.

Publications

- **The Illusion of Agentic Complexity in README.md Generation: Evaluating Single-Agent vs. Multi-Agent RAG Systems** *ICSME 2026 (Accepted), arXiv preprint*
- **Forecasting Finnish aFRR Energy Reserve Market Prices using Deep Learning and Tree-Based Models** *Elsevier Energy & AI, ScienceDirect*

Academic Research Projects

TinyTest: Small Language Models for Automated Unit Test Generation *GitHub Repository*

- Evaluated **Transformer-based SLMs** (Llama, Mistral) for unit-test generation via Z/few-shot prompting.
- Built an automated repair pipeline by feeding error logs back into context to mimic an **agentic feedback loop**.
- **Associated Skills:** Test Automation, Coverage, Mutmut, Python, Automated Evaluation, LLMs/SLMs.

Programming and Awards

- ICPC Regional: **Ranked 31st out of 1600** teams at the ICPC Regional Dhaka, placing in the top ranks.
- Google Kickstart: Ranked 29th in Bangladesh and **932nd globally** out of more than 7000 participants.
- Rankings: Achieved **Top 5** (CodeChef, Peak 2049) and **Top 15** (Codeforces, Peak 1607) national rankings.
- Problem Solving: Solved 3,000+ problems online and **authored 15 algorithmic challenges** (Graphs, DP).

Extracurricular Experiences

- **Judging Director, DIU Take-OFF (2023):** Led judging and problem curation for over 2,400 participants.
- **Problem Setter, NGPC (2022):** Designed algorithmic challenges for Bangladesh's leading women's contest.
- **VP, ACM Task Force (2022):** Organized national contest selections and mentored juniors in algorithms.
- **Technical Writer (2022):** Authored solution editorials for the Canadian Computing Competition.

References

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